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# Measuring Systemic Risk and Tail-Dependence in the Cryptocurrency Market using Forward-CoVaR

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# Introduction

The cryptocurrency market has grown from a small experiment into a huge global financial sector. It now features complex derivative markets, decentralized protocols, and massive flows of capital. However, the system is very fragile. We have seen this clearly during cascading failure events like the collapse of the FTX exchange and the collapse of the TerraUSD stablecoin. These events show how important it is to monitor and measure systemic risk in digital assets (Naifar, 2025). Systemic risk is the vulnerability of the whole financial sector. It is fundamentally different from the risk of a single asset because it captures the danger of a single failure starting a massive domino effect (Adrian and Brunnermeier, 2016). In the cryptocurrency world, these connections are very deep (Ji et al., 2019; Ferreira and Pereira, 2019). Exchanges, liquidity providers, and decentralized finance protocols are strongly connected through shared liquidity pools, stablecoins, and cross chain bridges.

A severe example of this systemic fragility is the threat that problems in one major platform can cause huge losses across all cryptocurrencies due to heavy contagion effects (Xu et al., 2021). The theoretical literature describes these mechanisms of joint distress very well. However, empirical methods usually fall back on static and isolated risk metrics that do not actually put these theories into practice. Current risk management relies heavily on standard metrics like Value at Risk or Expected Shortfall. These metrics only measure the downside exposure of a single asset all by itself (Fang et al., 2023). This big disconnect between identifying structural vulnerabilities in the literature and actual empirical forecasting completely ignores the economic reality of tail dependence (Borri, 2019; Waltz et al., 2022).

Ironically, the very mechanisms that are supposed to improve stability for individual assets can actually increase risk for the whole system. For example, many decentralized finance protocols use automatic liquidation rules and strict over collateralization requirements to protect lenders. These mechanisms do reduce the risk of individual defaults under normal conditions. But they can also heavily increase market stress during extreme events. Forced asset liquidations push prices down even further. This creates a negative feedback loop where distress spreads rapidly across the network (Naifar, 2025). Current unconditional risk models fail to capture these extreme tail events. As a result, capital buffers are completely insufficient right when market panics happen (Franco and Laurini, 2025).

To solve this main problem, this study provides a comprehensive synthesis to bridge the gap between the theoretical literature on financial contagion and the empirical methods needed to model cryptocurrency systemic risk. This analysis breaks down the transition from current risk measurement to predictive forecasting. It explains how abstract financial concepts like the traditional market to book ratio or macroeconomic liquidity stress are translated mathematically and empirically into blockchain native state variables. The following sections detail the econometric frameworks required to capture dynamic tail risk. They also explain how to use on chain metrics as a digital balance sheet, how to handle structural breaks caused by protocol upgrades, and the backtesting procedures needed to validate complex forecasting models.

# Literature Review

Many studies look at the risk and volatility of individual cryptocurrencies. However, there are far fewer studies that look at how distress spreads across the wider crypto market. In particular, less attention has been given to the mechanisms behind joint crashes and systemic contagion between different digital assets. To highlight this gap, this section first discusses the key literature on tail dependence, network interconnectedness, and how systemic risk has been measured over time. After establishing this theoretical background, the review then shifts to predictive frameworks, with the aim of identifying which on-chain metrics are most useful for forecasting future market stress.

## Systemic Tail Risk

To understand the new methods needed for digital assets, we must first look at how risk measurement evolved in traditional finance. The foundation of modern market risk management is Value at Risk. This is the maximum expected loss over a specific time period at a certain confidence level. Mathematically, the Value at Risk of an asset

i at quantile q is the threshold where the probability of a return $r_t^i$ falling below it equals $q$:

$$Pr(r_t^i \leq VaR_{q,t}^i) = q$$

Value at Risk is widely used by regulators. However, researchers agree it has a major flaw for systemic risk. It cannot measure the chance that many assets will pass their risk thresholds at the exact same time. Also, Value at Risk fails the property of subadditivity. This means the calculated risk of a portfolio could theoretically be higher than the sum of the risks of its individual parts. This makes it an inconsistent risk measure during extreme stress. To address the huge losses beyond the Value at Risk threshold, researchers introduced Expected Shortfall, which is also called Conditional Value at Risk. Expected Shortfall calculates the expected return on the days when the Value at Risk threshold is broken. This is an improvement for individual asset risk, but it still measures risk in isolation. To bridge the gap toward system wide assessment, Acharya, Pedersen, Philippon, and Richardson (2010) proposed Marginal Expected Shortfall. This measures the expected return of an individual firm when the whole market experiences an extreme crash.

The foundational framework for systemic risk analysis is the Conditional Value at Risk methodology developed by Adrian and Brunnermeier (2016) for the banking sector. They showed that an institutions contribution to systemic risk ($\Delta$ CoVaR) is often not correlated with its individual risk. Girardi and Ergün (2013) improved this methodology. They introduced a multivariate GARCH approach to estimate Conditional Value at Risk, which allows for dynamic time varying systemic risk measurement. To overcome the limitations of normal Gaussian assumptions in tail risk measurement, researchers have introduced advanced frameworks. Waltz et al. (2022) propose a Vulnerability Conditional Value at Risk estimated via copulas to capture asymmetric domino effects. González-Sánchez and Nave Pineda (2023) and Bruhn and Ernst (2022) also do this by using GARCH EVT Copula approaches. Alongside robust quantile regression frameworks, Zhao (2022) has also advanced CoVaR estimation through Extreme Value Theory (EVT) and dynamic copula approaches to explicitly capture severe, asymmetric tail dependence during market crashes.

## Systemic risk in cryptocurrency markets

In the cryptocurrency field, Borri (2019) was among the first to use conditional tail risk frameworks. He found high tail dependence among cryptocurrencies but low dependence with traditional financial markets. Building on this, many new studies focus on contagion and spillover effects. For example, Ferreira and Pereira (2019) and Ji et al. (2019) document the dynamic connections across different crypto assets. Furthermore, Bouri et al. (2021) and Xu et al. (2021) show that spillovers are very strong in the extreme quantiles of return distributions. This highlights the importance of tail risk dynamics during market distress.

Recent literature increasingly focuses on specific parts of the crypto market to better map these systemic vulnerabilities. Gunay et al. (2023) provide a sector analysis of systemic risk. Naifar (2025) maps tail risk specifically across decentralized finance, stablecoins, and infrastructure tokens. Łęt et al. (2023) explore the role of stablecoins as a potential stabilizing or destabilizing force. Extreme market events have also served as natural experiments for these models. For instance, Akhtaruzzaman et al. (2022) use network models to show increased systemic risk sharing and contagion during the Covid 19 pandemic. Also, Franco and Laurini (2025) highlight the need to capture rapid market shifts. They use a high frequency intraday approach to quantify systemic risk in crypto markets. This shows that systemic risk happens on much shorter time scales than traditional models suggest.

## Forward- $\Delta$ CoVar

A major criticism in the literature regarding standard systemic risk measures is that they are fundamentally contemporary measures. They describe how risk transmits in the present moment. This limits their usefulness for proactive risk management or macroprudential regulation. The literature notes that systemic risk typically builds up silently during periods of low volatility and high credit expansion. This phenomenon is known as the volatility paradox. By the time current metrics spike, the systemic event is already happening. This makes countercyclical interventions completely ineffective (Bartram et al., 2005; Hanif et al., 2020).

To bridge the gap between looking at the past and predicting the future, the methodology must include predictive

modeling. Adrian and Brunnermeier (2016) introduced the Forward- $\Delta$ CoVar method for the traditional banking sector. They achieved this by projecting current systemic risk contributions onto a set of past, firm specific balance sheet characteristics like size and leverage, alongside broad macroeconomic state variables.

By lagging the independent variables by a specific time horizon, the methodology shifts from describing current conditions to forecasting future vulnerability. The Forward-CoVaR framework theoretically assumes that structural imbalances inside an institution's balance sheet act as the catalyst for future systemic contagion. When an exogenous shock occurs, these existing structural imbalances determine how severe the subsequent spillover will be. Crucially, their empirical analysis demonstrated that structural imbalances such as a bank's leverage, size, and maturity mismatch are highly significant predictors of future systemic risk.

Their Forward $\Delta$ CoVaR measure was estimated during the quiet period of late 2006 and they successfully predicted more than a third of the realized systemic risk of the 2008 financial crisis. If these specific financial characteristics could signal the 2008 collapse well in advance, it justifies the attempt to find similar indicators within the cryptocurrency ecosystem.

Therefore, by measuring these imbalances today, we can predict which specific cryptocurrencies will contribute the most to systemic tail risk during future periods of distress. The challenge is to adapt this traditional banking methodology to the decentralized and cryptographic architecture of digital assets. Because cryptocurrency networks do not produce quarterly corporate balance sheets, it is necessary to create a digital balance sheet. Which includes on chain metrics and derivative market data that act as direct proxies for these significant predictors of systemic risk.

## Forecasting for cryptocurrencies using on chain indicators

Another part of the literature extends traditional risk measurement into predictive forecasting, marking a necessary shift from purely econometric volatility models to advanced analytics using machine learning and blockchain specific data.

Traditionally, forecasting Value at Risk and Expected Shortfall has relied heavily on variations of the GARCH model. For instance, Fang et al. (2023) use DCC GARCH models to forecast and backtest systemic risk in cryptocurrencies, while Chen and Liang (2024) and Fiszeder et al. (2024) emphasize the critical role of probability distribution choices and range based estimations to robustly forecast volatility. However, these traditional econometric models rely exclusively on historical price and trading volume. Consequently, they treat the asset largely as a black box and fail to capture the underlying technological utility, structural liquidity, and fundamental economic health of the network.

Because cryptocurrencies operate on public, transparent ledgers, they generate a wealth of real time, fundamental data that traditional financial markets lack. To overcome the limitations of price only models, recent literature has shifted toward integrating this blockchain native data, often utilizing machine learning frameworks capable of processing complex, non linear network dynamics. Researchers have increasingly demonstrated that specific on chain indicators possess significant predictive power that standard market data fails to capture.

Specifically, previous studies have successfully integrated and validated several key on chain metrics as empirical determinants of market behavior and risk. Network vitality metrics, such as active addresses and transaction throughput, contain crucial predictive information for trading strategies and network adoption that standard price histories completely miss, as demonstrated by King et al. (2024) and Pang et al. (2025). Furthermore, Zhang et al. (2026) provide evidence for the use of Total Value Locked as a core indicator of decentralized finance centrality. They demonstrate that Total Value Locked dynamics serve as the primary transmission channel for systemic liquidity shocks.

To process these multidimensional inputs, Wang et al. (2024) employ machine learning techniques such as extreme gradient boosting and neural networks. They find that incorporating network connection metrics significantly improves the prediction accuracy of systemic risk compared to traditional baselines. Solidifying this transition, Lee (2025) propose a machine learning based trading and forecasting strategy that actively combines technical indicators with fundamental on chain data. By specifically utilizing Total Value Locked and active addresses, they

demonstrate that integrating these blockchain native features significantly enhances the predictive accuracy of time varying models.

By highlighting these specific applications, the current literature establishes that on chain metrics are not merely descriptive statistics, but proven leading indicators required for robust, forward looking risk assessment.

### Identification of the Literature Gap

While existing literature extensively measures current contagion and details how risk spreads across the cryptocurrency market right now, there is still a significant gap in predictive modeling that uses blockchain native data. Most previous studies rely heavily on historical price and volume data. Consequently, very few researchers have analyzed how on chain health vitals, such as Hashrate, Total Value Locked, or Smart Contract activity, can be used to predict future systemic risk contributions. This approach is very similar to how Adrian and Brunnermeier (2016) used traditional balance sheet data to forecast systemic risk in the banking sector. To address this gap, this thesis treats on chain metrics as the digital equivalent of balance sheet characteristics. This offers a new predictive framework for digital assets.

Building on this identified gap, the main goal of this study is to analyze the predictive power of on chain data in forecasting the systemic risk contribution of major cryptocurrencies. Therefore, the main research question addressed in this thesis is whether on chain network characteristics can predict the future systemic risk contribution of cryptocurrencies more accurately than traditional market data alone. To thoroughly investigate this core problem, the study will also explore two important sub questions. First, it will examine how the ranking of systemically important coins changes when moving from an unconditional and static model to a conditional and time varying model. Second, it will identify which specific on chain variables, ranging from network Hashrate to decentralized liquidity measures, act as the strongest determinants of tail risk contagion.

Based on existing literature regarding financial contagion and network theory, this thesis tests three main hypotheses:

**H1** Cryptocurrencies show asymmetric tail dependence. This means that asset correlations are significantly stronger during severe market crashes compared to bullish periods (Borri, 2019).

**H2** Assets showing higher decentralized finance centrality, indicated by elevated Total Value Locked, will generate a higher systemic risk contribution.

**H3** On chain stress indicators, such as a sudden drop in Hashrate or a rapid spike in stablecoin outflows, serve as reliable leading indicators that statistically predict later increases in future systemic risk

## Data and Methodology

### Data Description

This study uses publicly available cryptocurrency market and blockchain data. The empirical analysis is based on a panel dataset covering the period from January 2020 to December 2025. This sample period captures multiple market regimes, including the expansionary phase of the recent bull market, subsequent deleveraging episodes, and prolonged bear market conditions, thereby providing a suitable environment for analysing systemic risk dynamics across different market states.

The empirical analysis focuses on five major cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Solana (SOL), and Binance Coin (BNB). Stablecoins are excluded due to their fundamentally different return dynamics, which may distort quantile-based systemic risk estimation. Restricting the sample to non-pegged assets ensures comparability of return distributions and tail behaviour.

The selected cryptocurrencies were chosen based on their market capitalisation, economic prominence, and functional importance within the digital asset ecosystem. Bitcoin (BTC) and Ethereum (ETH) represent the most established and liquid cryptocurrencies, playing a dominant role in price discovery and overall market dynamics. Ripple (XRP), Solana (SOL), and Binance Coin (BNB) are included due to their substantial market presence and their relevance across applications such as smart contract execution, decentralised finance (DeFi), exchange-based ecosystems, and

cross-border payment infrastructure. Together, these assets provide cross-sectional heterogeneity while preserving economic interpretability.

Market data consist of daily prices, returns, and trading activity measures, primarily sourced from CoinMetrics. System-wide risk is defined using a market-capitalisation-weighted system index constructed from the selected cryptocurrency universe. Consistent with the CoVaR framework of (Adrian and Brunnermeier, 2016), the system index excludes the cryptocurrency under analysis to avoid mechanical self-inclusion effects. The system index represents the aggregate cryptocurrency market return and forms the basis for measuring conditional tail risk.

Cryptocurrency-specific characteristics include blockchain-native indicators capturing valuation pressure, network activity, liquidity conditions, and ecosystem importance. These variables include Total Value Locked (TVL) active addresses and metrics such as hashrate where applicable. Hashrate measures are interacted with consensus-mechanism indicators to account for structural differences across networks.

On-chain and derivative-market data are sourced from Coinglass, DefiLlama, and CoinMetrics.

## Systemic Risk Framework

To model the systemic risk contribution of cryptocurrencies, this study adopts the Quantile Regression framework proposed by Adrian and Brunnermeier (2016). This approach enables the direct estimation of tail risk at the 1% and 5% quantiles, allowing systemic risk to be analysed within the distributional extremes rather than through mean dynamics alone. Systemic risk is measured using the Conditional Value-at-Risk (CoVaR) concept, with $\Delta CoVaR$ capturing the marginal contribution of individual cryptocurrencies to system-wide tail risk.

## Definition of CoVaR

Firstly, I establish the definitions of our risk measures. Recall that Value at Risk is defined as the $q$ quantile. This means:

$$\Pr(X^i \leq VaR_q^i) = q \tag{1}$$

where $X^i$ is the return variable of cryptocurrency $i$ for which the Value at Risk is defined.

We denote by $CoVaR_q^{sys|i}$ the Value at Risk of the financial system conditional on some event $\mathbb{C}(X^i)$ of cryptocurrency $i$. That is, $CoVaR_q^{sys|i}$ is implicitly defined by the $q$ quantile of the conditional probability distribution:

$$\Pr\left(X^{sys} \leq CoVaR_q^{sys|\mathbb{C}(X^i)} \mid \mathbb{C}(X^i)\right) = q \tag{2}$$

We denote the contribution of cryptocurrency $i$ to the systemic risk of the market by measuring the difference between the system Conditional Value at Risk when cryptocurrency $i$ is in distress versus when it is operating normally. The systemic risk contribution is then defined as:

$$\Delta CoVaR_q^{sys|i} = CoVaR_q^{sys|X^i=VaR_q^i} - CoVaR_q^{sys|X^i=Median^i} \tag{3}$$

## Unconditional CoVaR Estimation

To actually estimate these theoretical definitions, we use quantile regression. As a baseline specification, static systemic risk is estimated at the 5 percent quantile without conditioning on market state variables.

We first estimate the unconditional Value at Risk of cryptocurrency $i$ by running a quantile regression of its returns $X_t^i$:

$$X_t^i = \alpha_q^i + \varepsilon_{q,t}^i \tag{4}$$

where the predicted value from this regression gives us the empirical unconditional Value at Risk.

Next, we estimate the Conditional Value at Risk of the system conditional on the returns of cryptocurrency $i$:

$$X_t^{sys} = \alpha_q^{sys|i} + \beta_q^{sys|i} X_t^i + \varepsilon_{q,t}^{sys|i} \tag{5}$$

The unconditional systemic risk measure is obtained by evaluating this conditional quantile of system returns at the distress level of cryptocurrency $i$. The marginal systemic risk contribution is then calculated as:

$$\Delta CoVaR_{q,t}^{sys|i} = \hat{\beta}_q^{sys|i}(VaR_{q,t}^i - Median_t^i) \tag{6}$$

## Conditional CoVaR Estimation

To capture time variation in systemic risk, CoVaR is conditioned on lagged market state variables. Following Adrian and Brunnermeier (2016), both the Value-at-Risk of individual cryptocurrencies and the Conditional Value-at-Risk of the system are modelled as functions of market-wide state variables.

We first estimate the Value at Risk of cryptocurrency $i$ conditional on lagged market conditions:

$$X_t^i = \alpha_q^i + \gamma_q^i M_{t-1} + \varepsilon_{q,t}^i \tag{7}$$

where $M_{t-1}$ is the vector of lagged market state variables.

Next, we run the quantile regression for the system returns conditional on both the lagged market variables and the asset returns:

$$X_t^{sys} = \alpha_q^{sys|i} + \gamma_q^{sys|i} M_{t-1} + \beta_q^{sys|i} X_t^i + \varepsilon_{q,t}^{sys|i} \tag{8}$$

This specification allows systemic risk to vary with current market conditions. The predicted time varying Conditional Value at Risk is calculated as:

$$CoVaR_{q,t}^{sys|i} = \hat{\alpha}_q^{sys|i} + \hat{\gamma}_q^{sys|i} M_{t-1} + \hat{\beta}_q^{sys|i} VaR_{q,t}^i \tag{9}$$

The marginal systemic risk contribution over time is then calculated explicitly as:

$$\Delta CoVaR_{q,t}^{sys|i} = \hat{\beta}_q^{sys|i}(VaR_{q,t}^i - Median_t^i) \tag{10}$$

## Forward CoVaR Estimation

The Forward CoVaR measure is created by projecting the estimated time varying systemic risk contributions onto lagged cryptocurrency characteristics and lagged market state variables. Specifically we estimate:

$$\Delta CoVaR_{q,t}^{sys|i} = \alpha^i + \gamma^i M_{t-h} + \beta^i Z_{t-h}^i + \varepsilon_t^i \tag{11}$$

where $Z_{t-h}^i$ represents the vector of lagged cryptocurrency specific characteristics like on chain metrics, $M_{t-h}$ captures lagged market wide state variables, and $h$ is the forecast horizon.

The fitted value from this regression is defined as the Forward $\Delta CoVaR$ measure. This represents the predicted systemic risk contribution conditional on information available at an earlier time. Predictive performance is assessed using the quantile loss function, which is also known as the pinball loss.

### Model Validation and Backtesting

To evaluate the adequacy of the Forward CoVaR forecasts, model validation is conducted using the Kupiec Proportion of Failures test and the Christoffersen independence test (Kupiec, 1995) (Christoffersen, 1998). These procedures assess the statistical consistency of tail risk forecasts by evaluating violation frequency and clustering behavior.

## State Variables and Network Characteristics

To model the time-varying nature of systemic risk within the Forward-CoVaR framework, this study constructs two sets of lagged controls: the Market State vector ($M_{t-1}$) and the Coin Characteristic vector ($X_{t-1}^i$). All variables are constructed from daily observations and are lagged by one period to ensure predictive validity.

### General Variable Construction

Cryptocurrency returns are computed as logarithmic price differences:

$$r_t^i = \log(P_t^i) - \log(P_{t-1}^i) \tag{12}$$

where $P_t^i$ denotes the closing price of cryptocurrency $i$ at time $t$.

System-wide returns are defined using a market-capitalisation-weighted system index constructed from the selected cryptocurrency universe. For each cryptocurrency $i$, the system index excludes the asset under analysis to prevent mechanical correlation:

$$r_t^{sys,-i} = \sum_{j \neq i} w_{j,t} r_t^j \tag{13}$$

where the respective weights $w_{j,t}$ are determined by relative market capitalisation:

$$w_{j,t} = \frac{MC_{j,t}}{\sum_{k \neq i} MC_{k,t}} \tag{14}$$

Variables exhibiting strong trends or scale effects are transformed to ensure stationarity. For strongly trending variables, log-growth is employed:

$$\Delta \log(X_t) = \log(X_t) - \log(X_{t-1}) \tag{15}$$

Ratio-based valuation indicators are modelled using deviations from unity to centre the metrics around their theoretical fair-value baselines:

$$AdjRatio_t = Ratio_t - 1 \tag{16}$$

## Market State Vector ($M_{t-1}$)

To capture time variation in systemic risk, the CoVaR framework conditions on a vector of market-wide state variables. In line with Adrian and Brunnermeier (2016), the market state vector is designed to capture systematic conditions that shift tail risk for all assets simultaneously, rather than coin-specific shocks. Because crypto markets lack standardised daily macro-financial indicators (such as credit spreads) that cleanly map into decentralised markets, this study uses Bitcoin as a market-wide proxy. Bitcoin is the dominant and most liquid cryptoasset and plays a central role in information aggregation and price discovery, making its return and realised volatility suitable measures of market trend and market volatility. This choice reduces endogeneity relative to using asset-specific state variables and mirrors the role played by broad market indicators in Adrian and Brunnermeier (2016) framework. The state vector incorporates five distinct proxies. Firstly, Market Volatility serves as a proxy for short-term market stress and uncertainty. Consistent with the methodology of the BitMEX historical volatility indices (.BVOL), it is calculated as the 30-day rolling standard deviation of daily Bitcoin log returns explicitly annualized using a 365-day scaling factor to reflect the continuous trading nature of cryptocurrency markets:

$$M_{t-1}^{vol} = \sqrt{365} * \sqrt{\frac{1}{29}\sum_{\tau=1}^{30}\left(R_{t-\tau}^{BTC} - \bar{R}^{BTC}\right)^2} \tag{17}$$

Second, the prevailing market trend is established via the rolling mean of Bitcoin log returns:

$$M_{t-1}^{trend} = \frac{1}{90}\sum_{\tau=1}^{90} R_{t-\tau}^{BTC} \tag{18}$$

Third, systemic liquidity stress is measured using the volume-weighted aggregated Bitcoin funding rate across major perpetual futures exchanges, capturing the cost of leverage in the system:

$$M_{t-1}^{fund} = FundingRate_{t-1}^{BTC} \tag{19}$$

Fourth, fiat liquidity flow captures base fiat liquidity expansion or contraction within the broader digital asset ecosystem. This is computed as the daily log-growth of the combined market capitalisation of the dominant stablecoins, USDT and USDC:

$$M_{t-1}^{fiat} = \ln(MC_{t-1}^{USDT} + MC_{t-1}^{USDC}) - \ln(MC_{t-2}^{USDT} + MC_{t-2}^{USDC}) \tag{20}$$

Finally, market activity and associated trading frictions are quantified using the natural logarithm of Bitcoin spot trading volume:

$$M_{t-1}^{volu} = \ln(Vol_{t-1}^{BTC}) \tag{21}$$

## Coin Characteristic Vector ($X_{t-1}^{i}$)

Cross-sectional variation in systemic risk contributions is explained using a suite of cryptocurrency-specific characteristics. These encompass market-based variables alongside blockchain-native indicators that capture structural vulnerabilities, network fundamentals, liquidity dynamics, and ecosystem importance. Specifically, leverage intensity measures the derivatives exposure relative to the spot market size:

$$Lev_{i,t-1} = \frac{OI_{i,t-1}}{MC_{i,t-1}} \tag{22}$$

Network vitality is evaluated through the log-growth in active addresses, which serves as a proxy for user adoption

and transactional demand:

$$NetAct_{i,t-1} = \ln(AA_{i,t-1}) - \ln(AA_{i,t-2}) \tag{23}$$

For smart-contract platforms, DeFi liquidity risk is assessed via the log-growth in Total Value Locked (TVL), reflecting the expansion or contraction of capital committed to decentralised applications:

$$Liq_{i,t-1} = \ln(TVL_{i,t-1}) - \ln(TVL_{i,t-2}) \tag{24}$$

Additionally, for Proof-of-Work (PoW) networks, network security is measured as the log-growth in hashrate interacted with a PoW indicator dummy, capturing variations in computational expenditure securing the ledger:

$$Sec_{i,t-1} = [\ln(HR_{i,t-1}) - \ln(HR_{i,t-2})] \times D_{i,t-1}^{PoW} \tag{25}$$

**Valuation characteristics**

A fundamental challenge in applying traditional financial econometrics to digital assets is the structural heterogeneity across blockchain architectures. To operationalise the Market-to-Book (M/B) equity ratio from the foundational CoVaR framework (Adrian and Brunnermeier, 2016), this methodology employs a approach to accurately map valuation metrics to underlying network architectures.

For base-layer networks (such as BTC, ETH, and XRP), where the chronological movement of individual coins can be computationally tracked via UTXO models or transparent account ledgers, valuation overextension is measured using the Market-Value-to-Realized-Value (MVRV) ratio. The Realized Value computes the aggregate cost basis of the network by pricing each unit of the asset at the time of its last on-chain movement, thereby serving as a direct, native equivalent to Book Value:

$$Val_{i,t-1}^{Ledger} = MVRV_{i,t-1} - 1 \tag{26}$$

Conversely, for natively high-throughput platforms driven by decentralised finance ecosystems (such as SOL and BNB), the continuous, automated execution of internal smart contracts renders the computation of Realized Value computationally prohibitive. Therefore, the Market-to-Book ratio is proxied using the ratio of Market Capitalisation (MC) to Total Value Locked (TVL). In this framework, TVL represents the quantifiable, hard collateral deposited within the network's financial architecture:

$$Val_{i,t-1}^{DeFi} = \frac{MC_{i,t-1}}{TVL_{i,t-1}} \tag{27}$$

By bifurcating the characteristic vector in this manner, the model successfully adapts the core systemic vulnerability constraint of valuation overextension across functionally incompatible blockchain environments.

While Total Value Locked (TVL) is utilised in both the liquidity ($\Delta lnTVL$) and valuation (MC/TVL) characteristics, multicollinearity is avoided as the former measures the stationary flow of capital while the latter measures the level-based structural premium.

## Structural Break Handling

Cryptocurrency markets and blockchain networks are characterised by frequent regime shifts, protocol upgrades, and structural changes. These events may induce structural breaks in both market dynamics and blockchain-native indicators, potentially affecting parameter stability and inference.

Several mechanisms may generate structural instability. First, consensus-mechanism transitions alter the economic interpretation of network variables. Most notably, Ethereum's transition from Proof-of-Work (PoW) to Proof-of-Stake (PoS) fundamentally changes the relevance of hashrate and related security metrics. Second, major market disruptions, such as large-scale deleveraging events or exchange failures, may alter dependence structures and risk

Table 1: Summary of State Variables and Cryptocurrency Characteristics

| **Variable Classification** | **Symbol** | **Proxy / Calculation** | **Applicable Assets** |
|---|---|---|---|
| **Panel A: Market State Vector (Systematic Risk Framework)** | | | |
| Market Volatility | $M_vol_{t-1}$ | 30-day rolling standard deviation of BTC log returns | System-wide |
| Market Trend | $M_trend_{t-1}$ | 90-day rolling average of BTC log returns | System-wide |
| Systemic Liquidity Stress | $M_fund_{t-1}$ | Aggregated BTC volume-weighted funding rate | System-wide |
| Fiat Liquidity Flow | $M_fiat_{t-1}$ | Daily log-growth of USDT and USDC Market Capitalisation ($\Delta \ln(MC^{USDT} + MC^{USDC})$) | System-wide |
| Market Activity & Friction | $M_volu_{t-1}$ | Natural logarithm of BTC daily spot trading volume ($\ln Vol^{BTC}$) | System-wide |
| **Panel B: Coin Characteristic Vector (Idiosyncratic Risk Framework)** | | | |
| Leverage Intensity | $Lev_{i,t-1}$ | Open Interest / Market Capitalisation ($OI_i/MC_i$) | All Assets |
| Network Vitality | $NetAct_{i,t-1}$ | Daily log-growth of Active Addresses ($\Delta \ln AA_i$) | All Assets |
| DeFi Liquidity Risk | $Liq_{i,t-1}$ | Daily log-growth of Total Value Locked ($\Delta \ln TVL_i$) | ETH, SOL, BNB |
| Network Security | $Sec_{i,t-1}$ | Log-growth of Hashrate interacted with PoW Dummy | BTC, ETH (pre-Merge) |
| Valuation (Ledger) | $Val_{i,t-1}^{Ledger}$ | Market-Value-to-Realized-Value minus one ($MVRV_i - 1$) | BTC, ETH (pre-Merge), XRP |
| Valuation (DeFi) | $Val_{i,t-1}^{DeFi}$ | Market Capitalisation / Total Value Locked ($MC_i/TVL_i$) | ETH, SOL, BNB |

*Notes:* This table summarises the state variables utilised in the Forward-CoVaR quantile regressions. All variables are lagged by one period ($t-1$) to ensure predictive validity.

transmission mechanisms. Third, blockchain-native indicators may exhibit measurement discontinuities arising from network upgrades or changes in data-provider methodologies.

To mitigate the impact of structural breaks, multiple stabilisation strategies are employed. Variables exhibiting deterministic trends or scale effects are modelled using logarithmic transformations and growth-rate specifications. These transformations reduce non-stationarity and improve comparability across market regimes.

Consensus-mechanism differences are explicitly captured using regime-specific indicator variables. In particular, hashrate measures are interacted with consensus-mechanism dummies to ensure that parameter estimates reflect economically meaningful relationships.

In addition, robustness checks are conducted across subsamples corresponding to distinct market regimes. This approach evaluates the stability of systemic risk relationships and ensures that empirical results are not driven by isolated structural events.

Together, these procedures mitigate parameter instability, reduce spurious inference, and enhance the robustness of systemic risk estimation.

### Expected Signs of Coefficients

The Forward-CoVaR framework allows the predictive content of cryptocurrency characteristics to be interpreted through their directional impact on future systemic risk contributions. The expected signs of the panel regression coefficients follow from established systemic risk theory and the economic interpretation of blockchain-native indicators.

Leverage intensity is expected to exhibit a positive relationship with systemic risk contributions. Elevated derivative exposure, measured as the ratio of Open Interest to Market Capitalisation, reflects increased speculative positioning. High leverage increases the probability of cascading liquidations during market downturns, thereby amplifying shock transmission mechanisms across the ecosystem.

Valuation indicators, which capture structural overextension within investor positioning, are similarly expected to yield positive coefficients. For native ledgers, elevated MVRV values indicate high unrealised profitability, increasing vulnerability to abrupt, coordinated profit-taking. For smart-contract platforms, a high Market Capitalisation to Total Value Locked (MC/TVL) ratio indicates that speculative market pricing has vastly outpaced the fundamental collateral deposited within the network. In both architectures, structural overvaluation is expected to increase systemic fragility.

Conversely, DeFi liquidity risk, proxied by the log-growth of Total Value Locked, is expected to exhibit a negative coefficient. Positive TVL growth reflects fundamental capital inflows and deepening liquidity, which act as a buffer against systemic shocks. A contraction in TVL, analogous to a traditional bank run, signals rapid capital flight and is expected to increase systemic risk contributions.

Network security is also expected to exhibit a negative relationship with systemic risk. For Proof-of-Work assets, sustained growth in network hashrate reflects robust hardware-level security and miner profitability, reducing the network's vulnerability to endogenous infrastructure shocks or miner capitulation events.

Finally, network vitality, measured by the growth rate of active addresses, is expected to exhibit ambiguous or regime-dependent effects. While steady baseline adoption reflects robust economic utility and network health (implying a negative, stabilising effect on systemic risk), extreme, parabolic fluctuations in active addresses often signal retail speculative mania, which may exacerbate systemic instability.

Overall, characteristics capturing leverage and valuation overextension are expected to amplify systemic risk, while characteristics capturing fundamental liquidity growth and hardware security are expected to serve as systemic mitigants.

# Results

## Validation

# Conclusion

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